# I. Heat transfer

## Symbology

| Symbol | Quantity | Unit |
|---|---|---|
| $\lambda$ , $k$ | Thermal conductivity | W/(m · K) |
| $\dot{q}$ | Heat flux | W/m$^2$ |
| $\dot{Q}$ | Heat flow | W |
| $\alpha$ | Heat transfer coefficient | W/(m$^2$ · K) |
| $a$ | Thermal diffusivity | m$^2$/s |
| $\nu$ | Kinematic viscosity | m$^2$/s |

## 1. Types of heat transfer

### 1.1 Heat conduction

Occurs with all materials when there is a temperature gradient.

**Steady-state conduction (1st Fourier's law):**

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x}$$

**Transient conduction (2nd Fourier's law 1D):**

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho c_p} \frac{\partial^2 T}{\partial x^2}$$

### 1.2 Convective heat transfer

This occurs in moving fluids.

**Newton's approach**:

$$\dot{q} = \alpha \left(T_1 - T_2\right) = \frac{\lambda_{\text{fluid}}}{\delta_{\text{th}}} \left(T_1 - T_2\right)$$

### 1.3 Radiation

Bodies with a temperature above 0 K emit thermal radiation.

## 2. Heat conduction and conductivity

### 2.1 Heat flux

$$\dot{q} = -\lambda \frac{\mathrm{d}T}{\mathrm{d}x} = k\Delta T = \frac{\lambda}{\delta}\Delta T$$

Procedure:

$$\vec{\dot{q}} \int_0^\delta dx = -\lambda \int_{T_1}^{T_2} dT \Longrightarrow \dot{q} = \frac{\lambda}{\delta}\left(T_1 - T_2\right)$$

### 2.2 Heat flow

$$\dot{Q} = \frac{\Delta Q}{\Delta t} = \dot{q}A = kA\Delta T = \frac{\lambda}{\delta}A\Delta T$$

### 2.3 Thermal conduction $k^{-1}$ and Thermal conductivity $k$

$$k_{\text{tot}}^{-1} = \sum_{i=1}^{n} \frac{\delta_i}{\lambda_i} \Longrightarrow k_{\text{tot}} = \frac{1}{k_{\text{tot}}^{-1}}$$

## 3. Heat conduction through a hollow cylinder

$$\dot{Q} = -\lambda A \frac{\mathrm{d}T}{\mathrm{d}r} = -\lambda 2\pi r l \frac{\mathrm{d}T}{\mathrm{d}r}$$

$$\boxed{\dot{Q} = 2\pi l \cdot \frac{T_0 - T_1}{\frac{1}{\lambda} \ln \frac{r_1}{r_0}} = 2\pi r_0 l k \Delta T}$$

With multilayered walls:

$$\dot{Q} = 2\pi l \cdot \frac{T_0 - T_n}{\sum_{i=1}^{n} \frac{1}{\lambda_i} \ln \frac{r_i}{r_{i-1}}} = \boxed{2\pi r_0 l k_0 (T_0 - T_n)}$$

### 3.1 Thermal conductivity $k$

$$k_0 = \frac{1}{r_0\left(\left(\frac{1}{\lambda_1}\ln\frac{r_1}{r_0}\right) + \left(\frac{1}{\lambda_2}\ln\frac{r_2}{r_1}\right) + \cdots + \left(\frac{1}{\lambda_n}\ln\frac{r_n}{r_{n-1}}\right)\right)}$$

$$k_0^{-1} = r_0 \sum_{i=1}^{n} \left(\frac{1}{\lambda_i} \ln \frac{r_i}{r_{i-1}}\right)$$

### 3.2 Thin-walled pipe approximated as a flat wall

$$\dot{Q}_T = \lambda A \frac{T_0 - T_1}{\delta} \quad ; \quad A = 2\pi l \left(r_0 + \frac{\delta}{2}\right)$$

$$\frac{\dot{Q}_T}{\dot{Q}_W} = \frac{\delta}{\left(r_0 + \frac{\delta}{2}\right) \ln\left(1 + \frac{\delta}{r_0}\right)} = \frac{\delta}{r_0} \frac{1}{\left(1 + \frac{\delta}{2r_0}\right) \ln\left(1 + \frac{\delta}{r_0}\right)}$$

## 4. Convective heat transfer

### 4.1 Reynolds number

It compares inertial to viscous forces.

$$\boxed{Re = \rho\, c \frac{L_{\text{char}}}{\eta} = c \frac{L_{\text{char}}}{\nu}}$$

$Re < 2300$: Laminar flow;
$Re \approx 2300$: Transition zone;
$Re > 2300$: Turbulent flow

### 4.2 Prandtl number

It compares momentum diffusivity to thermal diffusivity. Relates the relative thickness of the velocity and thermal boundary layers.

$$Pr = \frac{\nu}{a} = \frac{\delta_{\text{fluid}}}{\delta_{\text{th}}} = \frac{\eta c_p}{\lambda} \quad ; \quad a = \frac{\lambda}{\rho c_p} \quad ; \quad \nu = \frac{\eta}{\rho}$$

### 4.3 Nusselt number

Measures convective heat transfer relative to pure conduction across a boundary layer and the heat transfer coefficient $\alpha$.

$$Nu = \frac{\alpha \cdot L_{\text{char}}}{\lambda_{\text{fl}}} = \dot{q}\frac{L_{\text{char}}}{\lambda \Delta T_{\text{char}}}$$

$$\alpha = \frac{Nu \cdot \lambda}{L_{\text{char}}}$$

#### 4.3.1 Forced convection

$$\boxed{Nu = c \cdot Re^m \cdot Pr^n}$$

where $c$ is a constant.

For laminar flow over a flat plate:

$$Nu = 0.664 \cdot \sqrt{Re} \cdot \sqrt[3]{Pr}$$

For a flow in the transition zone (Tz):

$$Nu_{\text{Tz}} = \sqrt{Nu_{\text{lam}}^2 + Nu_{\text{turb}}^2}$$

### 4.4 Grashof number

It compares buoyancy to viscous forces.

$$Gr = g\beta\Delta T \frac{L^3}{\nu^2} \quad ; \quad g = 9.81 \text{ N/kg}$$

$$\beta\Delta T = \frac{\rho_\infty - \rho_0}{\rho_\infty} = \frac{\Delta\rho}{\rho_\infty}$$

$$Gr = g\frac{\rho_\infty - \rho_0}{\rho_\infty}\frac{L^3}{\nu^2}$$

#### 4.4.1 Natural convection

**Peclet number**

It compares advective to diffusive heat transport, i.e., the strength of convection relative to thermal diffusion.

$$Pe = RePr$$

**Rayleigh number**

It combines buoyancy driving and thermal diffusion, and it measures the intensity of natural convection.

$$Ra = GrPr$$

## 5. Heat exchanger

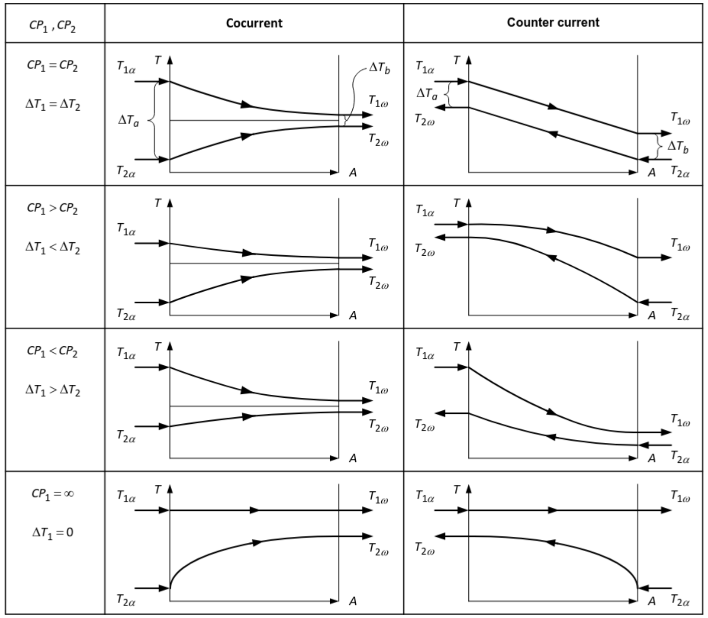
### 5.1 Energy balance

$$\dot{Q} = \dot{m}c_p\Delta\vartheta = CP\Delta\vartheta$$

$$\dot{Q}_1 = \dot{Q}_2$$

$$\dot{m}_1 c_{p1}\Delta\vartheta_1 = \dot{m}_2 c_{p2}\Delta\vartheta_2 \Longleftrightarrow CP_1\Delta\vartheta_1 = CP_2\Delta\vartheta_2$$

## 5.2 Average temperature difference




### 5.2.1 Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \frac{\Delta T_a}{\Delta T_b}}$$

where:

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

### 5.2.2 Mean temperature heat flow

$$\dot{Q} = kA\Delta T_m$$

### 5.2.3 Heat capacity rate ratio (Capacity ratio)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$CP_1\,(\vartheta_{1,\alpha} - \vartheta_{1,\omega}) = CP_2\,(\vartheta_{2\omega} - \vartheta_{2,\alpha})$$

$$R_1 = \frac{CP_1}{CP_2} = \frac{\vartheta_{2,\omega} - \vartheta_{2,\alpha}}{\vartheta_{1,\alpha} - \vartheta_{1,\omega}} = \frac{\Delta\vartheta_2}{\Delta\vartheta_1}$$

## 5.3 Co-current vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{\text{co}}\Delta T_{m,\text{co}} = kA_{\text{count}}\Delta T_{m,\text{count}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\dot{Q}_{\text{co}}}{\dot{Q}_{\text{count}}} \cdot \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}, \quad \text{if } \dot{Q}_1 \neq \dot{Q}_2$$

# II. The 2nd law of thermodynamics

## 6. Recap of the 1st law

## 6.1 Closed systems

$$q_{12} + w_{12} = u_2 - u_1$$

## 6.2 Open systems

$$q_{12} + w_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\,(z_2 - z_1)$$

$$w_{t12} = \int v\,dp + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\,(z_2 - z_1) + j_{12}$$

$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 \overset{\text{ideal gas}}{=\!=} c_p\,(T_2 - T_1)$$

## 7. The 2nd law

## 7.1 Statements

Energy = Exergy + Anergy

| Processes | Reversible | Irreversible |
|---|---|---|
| Exergy | Conserved | Reduced → Loss |
| Entropy | Constant | Increases → Production |

## 7.2 Entropy

Entropy $S$: extensive state variable [J/K]

Specific entropy $s$: intensive state variable [J/kgK]

Entropy flow $\dot{S}$: entropy over time [W/K]

## 8. Entropy balance equation

$$\boxed{dS = dS_Q + dS_m + dS_{irr}}$$

## 8.1 Entropy balance terms

- Heat transfer across the system boundary:

$$dS_Q = \frac{dQ}{T}$$

- Mass flow across the system boundary:

$$dS_{\text{m}} = \dot{m}\,s\;dt \quad ; \quad \dot{S}_m = \dot{m}\,s$$

- Irreversible processes inside the system:

$$\dot{S}_{irr}(t) \begin{cases} dS_{irr} > 0 & \text{irreversible processes} \\ dS_{irr} = 0 & \text{reversible processes} \\ dS_{irr} < 0 & \text{impossible} \end{cases}$$

## 8.2 Entropy balance equation for closed systems

$$dS = dS_Q + \cancelto{0}{dS_m} + dS_{irr}$$

$$dS = \dot{S}_Q\;dt + \dot{S}_{irr}\;dt = \left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right) dt$$

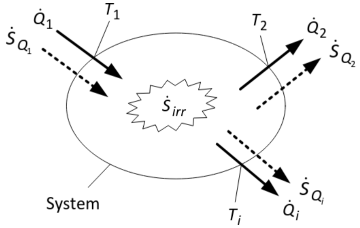
### 8.2.1 Closed system entropy flow balance equation

$$\frac{dS}{dt} = \sum_i \dot{S}_{Qi} + \dot{S}_{irr} \quad ; \quad \dot{S}_{irr} \geq 0$$

For steady-state case:

$$\dot{S}_{irr} = -\sum_i \dot{S}_{Qi} = -\sum_i \frac{\dot{Q}_i}{T_i} \geq 0$$

### 8.2.2 Visual representation




$$\boxed{\sum_i \dot{S}_{Qi} = \sum_i \frac{\dot{Q}_i}{T_i} = -\dot{S}_{irr}}$$

## 8.3 Entropy balance equation for open systems

$$\frac{dS}{dt} = \sum_\alpha \dot{m}_\alpha s_\alpha - \sum_\omega \dot{m}_\omega s_\omega + \sum \dot{S}_Q(t) + \dot{S}_{irr}(t)$$

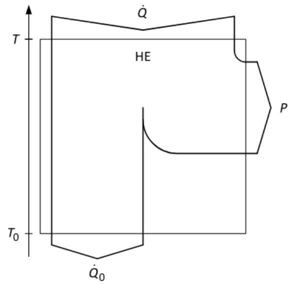
### 8.3.1 Steady-state flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha - \sum \dot{S}_Q \geq 0$$

### 8.3.2 Steady-state and adiabatic flow processes

$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha \geq 0$$

## 9. The thermal engine




$$\dot{Q} + \dot{Q}_0 + P = 0$$

$$\eta_{th} = \frac{-P}{\dot{Q}} = 1 - \frac{\left|\dot{Q}_0\right|}{\dot{Q}}$$

$$\dot{S}_{irr} = -\sum_{i=1}^{n} \dot{S}_{Qi} = -\frac{\dot{Q}}{T} - \frac{\dot{Q}_0}{T_0} \geq 0$$

$$-\dot{Q}_0 = T_0\left(\frac{\dot{Q}}{T} + \dot{S}_{irr}\right)$$

## 9.1 Carnot efficiency

$$\eta_{th} = 1 - \frac{T_0}{T} - \cancelto{0,\ \text{ideal}}{\frac{T_0\dot{S}_{irr}}{\dot{Q}}}$$

$$\eta_{th,max} = \eta_C(T, T_0) = 1 - \frac{T_0}{T}$$

# 10. Entropy $S$ as a state variable

$$ds = \frac{dq + dj}{T} = \frac{du + p\,dv}{T} = \frac{dh - v\,dp}{T}$$
$$T\,ds = dq + dj \implies T\,(s_2 - s_1) = q_{12} + j_{12}$$
$$\text{General form: } \int_1^2 T\,ds = q_{12} + j_{12}$$

## 10.1 Reversible processes

### 10.1.1 Clausius entropy relation

$$\boxed{\delta q_{\text{rev}} = T\,ds \iff \delta Q_{\text{rev}} = T\,dS}$$

### 10.1.2 Work $W$ and heat $Q$

$$w_{p,\,rev} = \int v\,dp$$
$$Q_{rev} = \int T\,dS$$

## 10.2 Dissipation $j$ as a state variable

### 10.2.1 Dissipation

$$w_{t12} = w_{p12} + j_{12} \iff w_{p12} = w_{t12} - j_{12}$$
$$dq_{rev} + dw_{v,\,rev} = du$$
$$T\,ds = dq + T\,ds_{irr} = dq_{rev} + dj$$
$$\int_1^2 T\,ds = q_{12} + j_{12}$$

### 10.2.2 For closed systems

$$ds = \frac{du + p\,dv}{T} = \frac{c_v\,dT}{T} + \frac{R_i\,dv}{v}$$
$$s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1}$$

### 10.2.3 Work change

All the works can be calculated with a temperature change:

$$q_{12} + w_{t12} = h_2 - h_1 + \frac{c_2^2 - c_1^2}{2} + g\,(z_2 - z_1)$$
$$q_{12} + \int v\,dp + j_{12} = h_2 - h_1 = c_p\,(T_2 - T_1)$$
$$\int T\,ds = q_{12} + j_{12} = \boxed{\int T\,ds + \int v\,dp = h_2 - h_1}$$
$$\boxed{T\,ds + v\,dp = h_2 - h_1 \iff T\,ds - p\,dv = u_2 - u_1}$$
$$T\,ds + v\,dp = dh\,T\,ds - p\,dv = du$$

## 10.3 Reversible state changes with perfect gases

### 10.3.1 Open system

$$q_{12} + j_{12} + w_{p,rev} = h_2 - h_1$$
$$\int_1^2 T\,ds + \int_1^2 v\,dp = c_p\,(T_2 - T_1)$$

### 10.3.2 Closed system

$$q_{12} + j_{12} + w_{v,rev} = u_2 - u_1$$
$$\int_1^2 T\,ds - \int_1^2 p\,dv = c_v\,(T_2 - T_1)$$

## 10.4 Polytropic state changes

$$T\,ds = dq + dj = du + p\,dv = dh - v\,dp$$
$$du + p\,dv = c_v\,dT + p\,dv \quad ; \quad dh - v\,dp = c_p\,dT - v\,dp$$
$$dw_v = c_v \frac{k-1}{n-1}\,dT = -p\,dv$$

For the change in entropy of the polytropic state change:

$$ds = c_v \frac{n-k}{n-1} \frac{dT}{T}$$
$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$
$$\boxed{s_2 - s_1 = c_v \frac{n-k}{n-1} \ln\frac{T_2}{T_1}} \quad n \neq 1$$
$$\boxed{s_2 - s_1 = c_v \ln\frac{T_2}{T_1} + R_i \ln\frac{v_2}{v_1} = c_p \ln\frac{T_2}{T_1} - R_i \ln\frac{p_2}{p_1}}$$

**Compressor with dissipation**: $n > k$
**Turbine with dissipation**: $n < k$

# 11. Exergy and anergy

**Exergy**: useful part of energy that can be completely converted into work under given ambient conditions.

**Anergy**: remaining part of energy that cannot be converted into work under given ambient conditions.

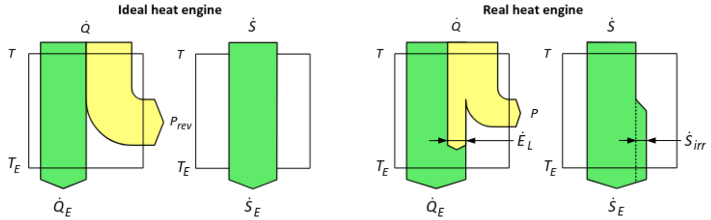
$$\boxed{e = ex + an}$$

## 11.1 Statements of the 2nd LT

**I.** In all irreversible processes exergy is converted into anergy
**II.** Only in reversible processes exergy remains constant
**III.** It is impossible to convert anergy into exergy

## 11.2 Exergy

### 11.2.1 Exergy balance / Exergy loss




$$\dot{E}_L = \sum_\alpha \dot{E}_\alpha - \sum_\omega |\dot{E}_\omega| = \dot{Q}\frac{T_A - T_E}{T_A} + \left(-\dot{Q}\frac{T_B - T_E}{T_B}\right)$$
$$\boxed{\dot{E}_L = T_E \dot{S}_{irr} = T_E \frac{T_A - T_B}{T_A T_B} \dot{Q}}$$
$$\dot{S}_{irr} = \sum_\omega \dot{m}_\omega s_\omega - \sum_\alpha \dot{m}_\alpha s_\alpha - \sum \dot{S}_Q \geq 0, \quad d\dot{S}_Q = \frac{d\dot{Q}}{T}$$

### 11.2.2 Exergy of heat

$$\dot{E}_Q = P_{rev} = \left(1 - \frac{T_E}{T}\right)\dot{Q} = \eta_C \dot{Q}$$

### 11.2.3 Exergy of material flows (without mixtures)

$$e = h - h_E - T_E\,(s - s_E) + \frac{c^2}{2} + gz + q\left(1 - \frac{T_E}{T}\right)$$
$$e_2 - e_1 = h_2 - h_1 - T_E\,(s_2 - s_1)$$

**Perfect gases**

$$e = c_p \left(T - T_E - T_E \ln\frac{T}{T_E}\right) + R_i T_E \ln\frac{p}{p_E}$$

**Incompressible liquids (isobaric)**

$$e = c_p \left(T - T_E - T_E \ln\frac{T}{T_E}\right)$$

**Isobaric for $T < T_E$ (cooling)**

$$e = c_p \left(T_E - T - T_E \ln\frac{T_E}{T}\right)$$

# III. Steady-state flow processes

## 12. Adiabatic working cycles

### 12.1 Compressor

#### 12.1.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} + q_{12} = c_p\left(T_2 - T_1\right) + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$P_{12} + \dot{Q}_{12} = \dot{m}\left(w_{t12} + q_{12}\right)$$

#### 12.1.2 Isentropic compression efficiency factor

$$\eta_s = \frac{\text{isentropic work}}{\text{real polytropic work}}$$

$$\eta_s = \frac{\Delta h_s}{\Delta h} = \frac{h_{2,s} - h_1}{h_2 - h_1} = \frac{c_{p,m}\left(T_{2,s} - T_1\right)}{c_{p,m}\left(T_2 - T_1\right)} = \frac{T_{2,s} - T_1}{T_2 - T_1} \leq 1$$

### 12.2 Turbine

#### 12.2.1 1st law

$$w_{t12} + q_{12} = h_2 - h_1 + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$w_{t12} + q_{12} = c_p\left(T_2 - T_1\right) + \frac{1}{2}\left(c_2^2 - c_1^2\right) + g\left(z_2 - z_1\right)$$

$$P_{12} = \dot{m} w_{t12} \iff -P_{12} = \dot{m}\left|w_{t12}\right|$$

#### 12.2.2 Isentropic turbine efficiency

$$\eta_s = \frac{\text{real polytropic work}}{\text{isentropic work}}$$

$$\eta_s = \frac{-\Delta h}{-\Delta h_s} = \frac{h_1 - h_2}{h_1 - h_{2,s}} = \frac{T_1 - T_2}{T_1 - T_{2,s}}$$

## 13. Adiabatic flow processes without work

### 13.1 Throttle / Expansion valve (isenthalpic)

$$h_1 = h_2 \implies j_{12} = -\int_1^2 v\,dp = -R_i T \ln\frac{p_2}{p_1}$$

### 13.2 Nozzle

$$h_2^0 = h_1^0 \implies h_2 + \frac{1}{2}c_2^2 = h_1 + \frac{1}{2}c_1^2$$

$$c_2 = \sqrt{2\left(h_1 - h_2\right) + c_1^2}$$

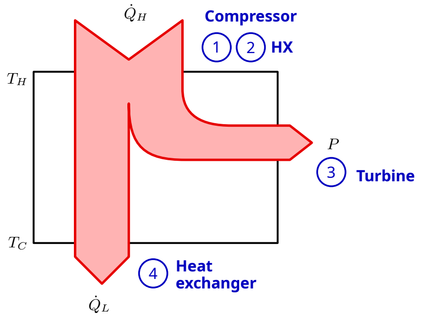
where $h^0$ is the total enthalpy: $h^0 = h + \frac{c^2}{2}$

# IV. Thermodynamics cycles

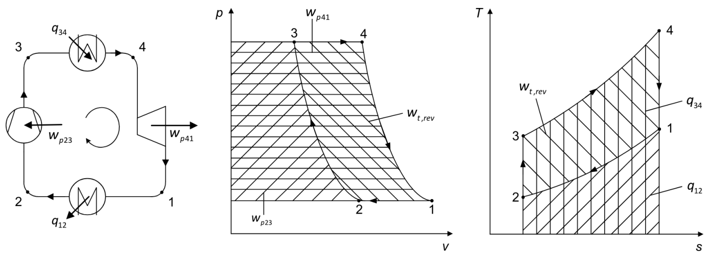
## 14. Reversible cycle work

Total cycle / sum of all sub-processes:

$$\boxed{w_{t,\,rev} = \int_1^2 v\,dp + \int_2^3 v\,dp + \ldots + \int_n^1 v\,dp = \oint v\,dp = \sum_{i=1}^{n} w_{pi}}$$



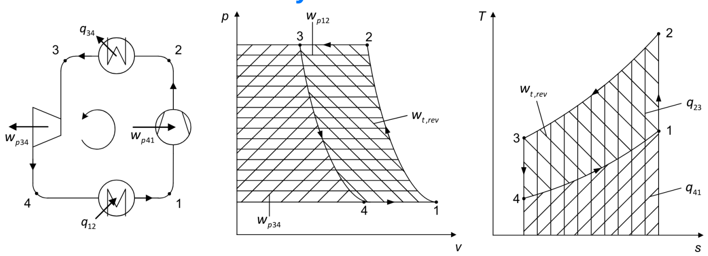

### 14.1 Clockwise cycle




$$w_{t,rev} = w_{p,23} + w_{p,41} < 0$$

$$\left|w_{t,rev}\right| = \left|q_{34}\right| - \left|q_{12}\right|$$

### 14.2 Counterclockwise cycle

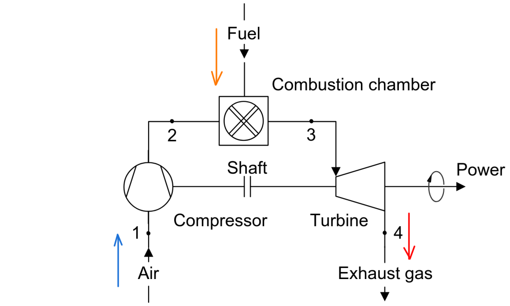


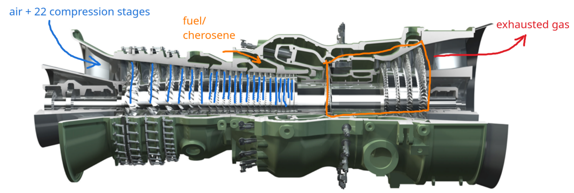

$$w_{t,rev} = w_{p,12} + w_{p,34} < 0$$

$$\left|w_{t,rev}\right| = \left|q_{23}\right| - \left|q_{41}\right|$$

# V. Thermal systems with gas turbines

## 15. Turbine




Power balance:

$$\dot{Q}_{23} + \dot{Q}_{41} + P_C = 0$$

### 15.1 Joule process

$$\pi = \frac{p_2}{p_1} = \frac{p_3}{p_4}$$

$$-w_C = q_\alpha + q_\omega = q_{23} + q_{41} = c_p\left(T_3 - T_2 + T_1 - T_4\right)$$

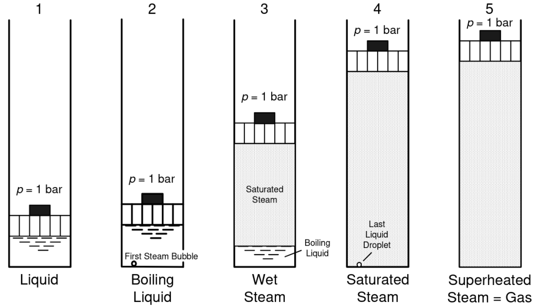
$$\left|w_C\right| = c_p\left(T_3 - T_2\right)\left(1 - \frac{T_4 - T_1}{T_3 - T_2}\right)$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{p}{p_0}\right)^{\frac{k-1}{k}} = \left(\frac{1}{\pi}\right)^{\frac{k-1}{k}}$$

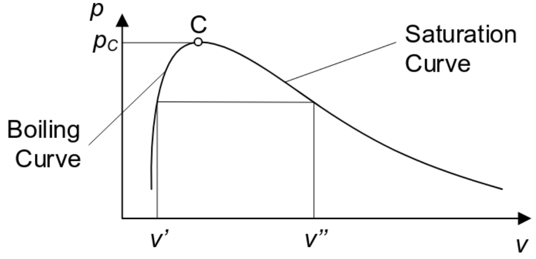
$$\eta_J = \frac{\left|w_C\right|}{q_\alpha} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{1}{\pi}\right)^{\frac{k-1}{k}}$$

# VI. Thermal plants with steam turbines

## 16. Types of steam




## 17. Boundary curves of the two phase region




### 17.1 Steam tables

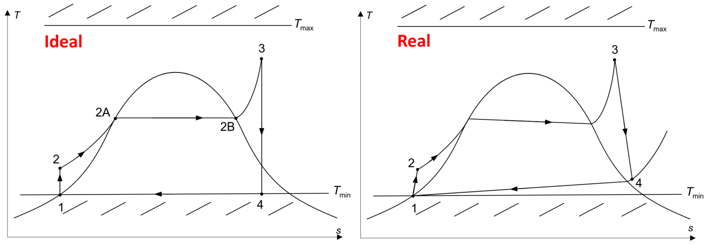
$$v = (1-x)v' + xv'' = v' + x(v'' - v')$$

$$h = (1-x)h' + xh'' = h' + x\Delta h_v$$

$$s = (1-x)s' + xs'' = s' + x\left(\frac{\Delta h_v}{T}\right)$$

## 18. Steam power stations

### 18.1 Rankine cycle



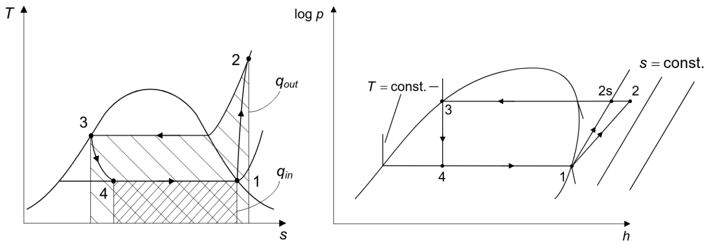

$w_{t12} = h_2 - h_1$ (pump) $\quad |w_{t34}| = h_3 - h_4$ (turbine)

$q_{23} = h_3 - h_2$ (boiler) $\quad |q_{41}| = h_4 - h_1$ (condenser)

$$\eta_{th} = \frac{(h_3 - h_4) - \overbrace{(h_2 - h_1)}^{\text{pump work}}}{h_3 - h_2} \quad ; \quad \eta_s = \frac{h_3 - h_4}{h_3 - h_{4,s}} \le 1$$

# VII. Heat pumps and Chillers

## 19. T-s and log p-h diagrams




$w_{t,12} = h_2 - h_1$ (compressor) $\quad h_3 = h_4$ (expansion valve)

$q_{23} = h_2 - h_3$ (condenser) $\quad q_{41} = h_1 - h_4$ (evaporator)

### 19.1 Isentropic efficiency (compressor)

$$\eta_s = \frac{h_{2,s} - h_1}{h_2 - h_1} \le 1$$

## 20. Coefficient of performance

### 20.1 Inner coefficient of performance

#### 20.1.1 Inner COP of a heat pump

$$\varepsilon_{\text{HP}} = \frac{|q_{23}|}{w_{12}} = \frac{\text{heat}}{\text{technical work}} = \frac{h_2 - h_3}{h_2 - h_1} = \varepsilon_R + 1$$

$$\text{Carnot: } \frac{1}{\eta_C} = \varepsilon_{HP,C} = \frac{T_C}{T_C - T_E} = \frac{T_C}{\Delta T_{\text{lift}}} = \varepsilon_{R,C} + 1$$

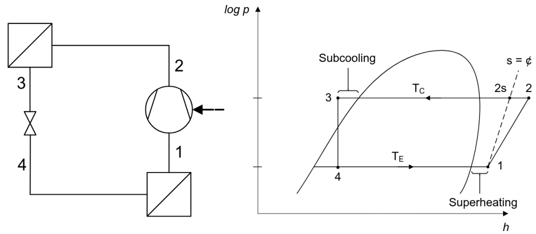
#### 20.1.2 Inner COP of a refrigerator

$$\varepsilon_R = \frac{|q_{41}|}{w_{12}} = \frac{h_1 - h_4}{h_2 - h_1} = \varepsilon_{\text{HP}} - 1$$

$$\text{Carnot: } \varepsilon_{R,C} = \frac{T_E}{T_C - T_E} = \frac{T_E}{\Delta T_{\text{lift}}} = \varepsilon_{HP,C} - 1$$

### 20.2 Outer coefficient of performance (COP)

#### 20.2.1 COP of a heat pump




$$\text{COP}_{\text{HP}} = \frac{|\dot{Q}_\omega|}{P_{el}} = \frac{|\dot{Q}|}{P_{el}}$$

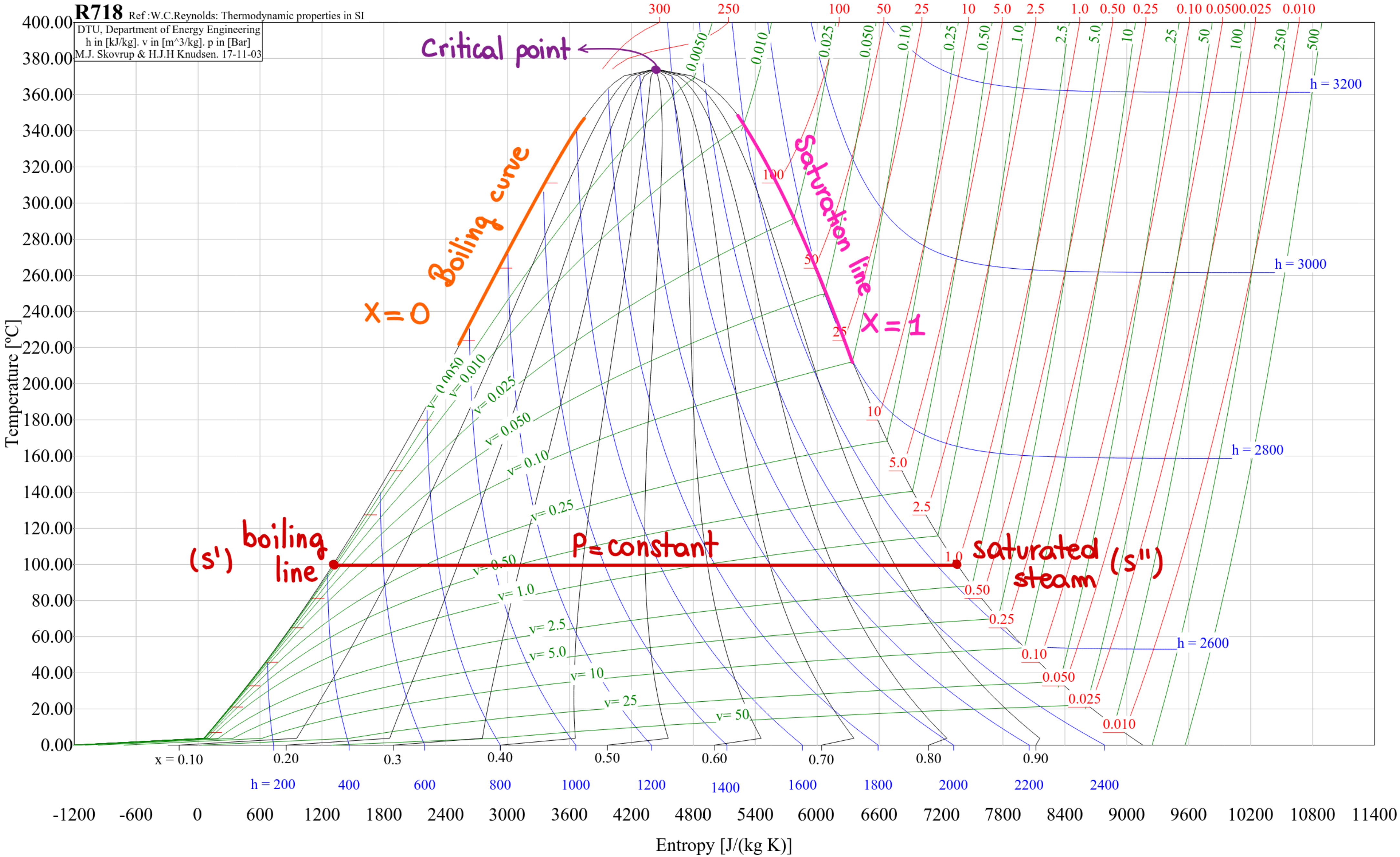
$$\varepsilon_{HP} > COP_{HP} \quad ; \quad \zeta_{HP} = \frac{COP_{HP}}{\varepsilon_{HP,C}}$$

#### 20.2.2 COP of a refrigerator

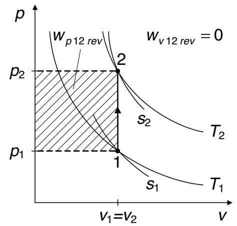
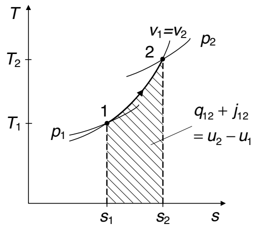
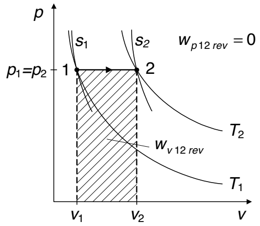
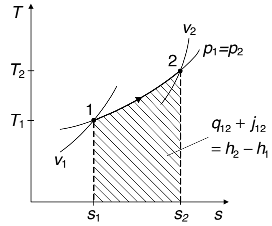
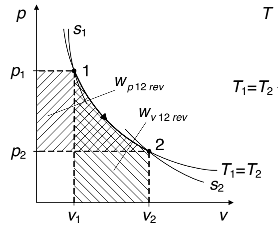
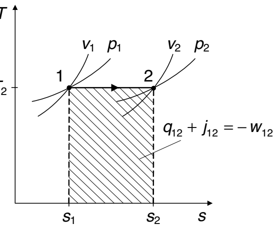
$$EER = COP_R = \frac{\dot{Q}_{in}}{P_{el}} = \frac{\dot{Q}_e}{P_{el}}$$

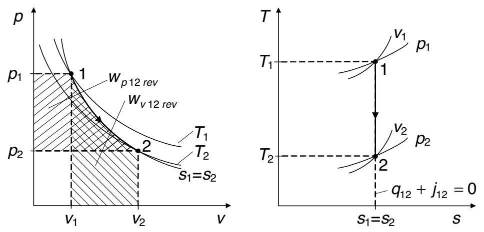
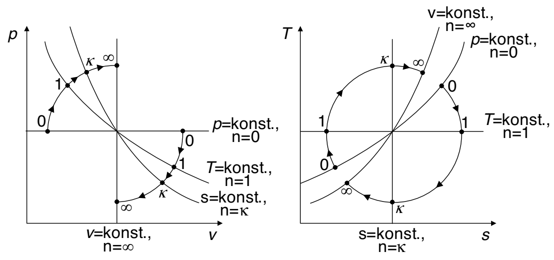
$$\varepsilon_R > EER \quad ; \quad \zeta_R = \frac{EER}{\varepsilon_{R,C}}$$

## A. T-s diagram


R718 Ref :W.C.Reynolds: Thermodynamic properties in SI
DTU, Department of Energy Engineering
h in [kJ/kg]. v in [m^3/kg]. p in [Bar]
M.J. Skovrup & H.J.H Knudsen. 17-11-03
Critical point
Boiling curve
X=0
Saturation line
X=1
(s')
boiling line
P=constant
Saturated steam
(s'')
Temperature [ºC]
Entropy [J/(kg K)]
h = 3200
h = 3000
h = 2800
h = 2600

## Special state changes with perfect gases (without consideration of kinetic and potential energies)

| **Ideal gas** | $v=$ **constant: Isochoric state change** | $p=$ **constant: Isobaric state change** | $T=$ **constant: Isothermal state change** |
|---|---|---|---|
| **Therm. state equation**<br>$pV = n\,R\,T$<br>$R = 8.314\,\mathrm{J/mol\,K}$<br>$pV = m\,R_i\,T$<br>$R_i = \dfrac{R}{M_i}$<br>$p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$<br>$R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$<br>$c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$ | <br>$v = \text{const.:} \quad \dfrac{p_1}{T_1} = \dfrac{p_2}{T_2} = \dfrac{p}{T} = \text{const.}$<br>$w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$<br>$w_{p\,12\,rev} = v\,(p_2 - p_1) = \dfrac{p_2 - p_1}{\rho}$ | <br>$p = \text{const.:} \quad \dfrac{v_1}{T_1} = \dfrac{v_2}{T_2} = \dfrac{v}{T} = \text{const.}$<br>$w_{v\,12\,rev} = -p\,(v_2 - v_1) = -R_i\,(T_2 - T_1)$<br>$w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$ | <br>$T = \text{const.:} \quad p_1\,v_1 = p_2\,v_2 = p\,v = \text{const.}$<br>$w_{v\,12\,rev} = -p_1\,v_1\,\ln\dfrac{v_2}{v_1} = \underbrace{p_1\,v_1\,\ln\dfrac{p_2}{p_1}}_{[J]} = \underbrace{R_i\,T\,\ln\dfrac{p_2}{p_1}}_{[J/kg]}$<br>$w_{p\,12\,rev} = w_{v\,12\,rev}$ |
| **1st LT closed system**<br>$q_{12} + w_{12} = u_2 - u_1$<br>Rev. volume change work:<br>$w_{v\,12\,rev} = -\displaystyle\int_1^2 p\,\mathrm{d}v$<br>$w_{v\,12} = w_{v\,12\,rev} + j_{12}$<br>Overall work:<br>$w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$ | $q_{12} + w_{12} = u_2 - u_1$<br>Rev. SC:<br>$w_{12} = w_{v\,12\,rev} = 0 \quad (\mathrm{d}v = 0)$<br>$q_{12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$ | $q_{12} + w_{12} = u_2 - u_1$<br>Rev. SC:<br>$w_{12} = w_{v\,12\,rev} = -p\,(v_2 - v_1)$<br>$q_{12\,rev} = u_2 - u_1 + p\,(v_2 - v_1) = h_2 - h_1$<br>$q_{12\,rev} = c_p\,(T_2 - T_1)$ | $q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1) = 0$<br>$q_{12} = -w_{12}$<br>Rev. SC:<br>$q_{12\,rev} = -w_{v\,12\,rev}$<br>$q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$ |
| **1st LT open system**<br>$q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. pressure change work:<br>$w_{p\,12\,rev} = \displaystyle\int_1^2 v\,\mathrm{d}p$<br>Technical work:<br>$w_{t\,12} = \displaystyle\int_1^2 v\,\mathrm{d}p + j_{12}$ | $q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. SC:<br>$w_{t\,12} = w_{p\,12\,rev} = v\,(p_2 - p_1)$<br>$q_{12\,rev} = h_2 - h_1 - v\,(p_2 - p_1) = u_2 - u_1$<br>$q_{12\,rev} = c_v\,(T_2 - T_1)$ | $q_{12} + w_{t\,12} = h_2 - h_1$<br>Rev. SC:<br>$w_{t\,12} = w_{p\,12\,rev} = 0 \quad (\mathrm{d}p = 0)$<br>$q_{12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$ | $q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1) = 0$<br>$q_{12} = -w_{t\,12}$<br>Rev. SC:<br>$q_{12\,rev} = -w_{p\,12\,rev}$<br>$q_{12\,rev} = p_1\,v_1\,\ln\dfrac{v_2}{v_1} = -p_1\,v_1\,\ln\dfrac{p_2}{p_1} = -R_i\,T\,\ln\dfrac{p_2}{p_1}$ |
| **2nd LT / entropy**<br>$\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$<br>$\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$ | $\mathrm{d}s = \dfrac{\mathrm{d}u}{T} = \dfrac{c_v\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1}$ | $\mathrm{d}s = \dfrac{\mathrm{d}h}{T} = \dfrac{c_p\,\mathrm{d}T}{T} \Rightarrow s_2 - s_1 = c_p\,\ln\dfrac{T_2}{T_1}$ | $\mathrm{d}s = \dfrac{p\,\mathrm{d}v}{T} = \dfrac{R_i\,\mathrm{d}v}{v} \Rightarrow s_2 - s_1 = R_i\,\ln\dfrac{v_2}{v_1}$<br>$\mathrm{d}s = -\dfrac{v\,\mathrm{d}p}{T} = -\dfrac{R_i\,\mathrm{d}p}{p} \Rightarrow s_2 - s_1 = -R_i\,\ln\dfrac{p_2}{p_1}$ |

| Ideal gas | $s=$ constant: Isentropic state change | $n=$ constant: Polytropic state change |
|---|---|---|
| **Therm. state change**<br>$pV=n\,R\,T$<br>$R=8.314\,\mathrm{J/mol\,K}$<br>$pV=m\,R_i\,T$<br>$R_i=\frac{R}{M_i}$<br>$pv=R_i\,T,\quad \rho=\frac{p}{R_i\,T}$<br>$R_i=c_p-c_v,\quad \kappa=\frac{c_p}{c_v}$<br>$c_v=\frac{R_i}{\kappa-1},\quad c_p=\frac{\kappa\,R_i}{\kappa-1}$ | <br>$s=\text{const}:\quad p_1\,v_1^\kappa=p_2\,v_2^\kappa=\text{const.}\quad\Rightarrow\quad \frac{T_2}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}}=\left(\frac{v_1}{v_2}\right)^{\kappa-1}$<br>$w_{v\,12\,rev}=\frac{p_1\,v_1}{\kappa-1}\left[\left(\frac{v_1}{v_2}\right)^{\kappa-1}-1\right]=$ etc. (replace)<br>$w_{p\,12\,rev}=\kappa\,w_{v\,12\,rev}=\kappa\,\frac{p_1\,v_1}{\kappa-1}\left[\left(\frac{v_1}{v_2}\right)^{\kappa-1}-1\right]=$ etc. (replace) | <br>Polytrope: $pv^n=$ konst.<br>Isobare: n=0, $pv^0=$ konst.<br>Isotherme: n=1, pv = konst.<br>Isentrope: n=κ, $pv^\kappa=$ konst.<br>Isochore: n=∞, $pv^\infty=$ konst.<br>$n=\frac{\ln\frac{p_2}{p_1}}{\ln\frac{v_1}{v_2}}=\frac{\ln\frac{p_2}{p_1}}{\ln\frac{p_2}{p_1}-\ln\frac{T_2}{T_1}}$<br>$n=\text{const}:\quad p_1\,v_1^n=p_2\,v_2^n=\text{const.}\quad\Rightarrow\quad \frac{T_2}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}=\left(\frac{v_1}{v_2}\right)^{n-1}$<br>$w_{v\,12}=\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. (replace, $n\neq1$)<br>$w_{p\,12}=n\,w_{v\,12}=n\,\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. (replace, $n\neq1$) |
| **1st LT closed system**<br>$q_{12}+w_{12}=u_2-u_1$<br>Rev. volume change work:<br>$w_{v\,12\,rev}=-\int_1^2 p\,\mathrm{d}v$<br>$w_{v\,12}=w_{v\,12\,rev}+j_{12}$<br>Overall work:<br>$w_{12}=w_{v\,12}+w_{W\,12}+w_{el\,12}$ | $q_{12}+w_{12}=u_2-u_1$<br>Isentropic means adiabatic ($q_{12}=0$) and reversible ($j_{12}=0$):<br>$w_{12}=w_{v12\,rev}=u_2-u_1=c_v\,(T_2-T_1)$<br>$w_{v12\,rev}=\frac{p_1\,v_1}{\kappa-1}\left[\left(\frac{v_1}{v_2}\right)^{\kappa-1}-1\right]=\frac{R_i\,T_1}{\kappa-1}\left[\left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}}-1\right]=$ etc. | $q_{12}+w_{12}=u_2-u_1=c_v\,(T_2-T_1)$<br>$w_{12}=w_{v\,12}=\frac{R_i}{n-1}\,(T_2-T_1)=c_v\,\frac{\kappa-1}{n-1}\,(T_2-T_1)=$<br>$=\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. $\quad(n\neq1)$<br>Rev. SC:<br>$w_{12}=w_{v12\,rev}$<br>$q_{12\,rev}=c_v\,\frac{n-\kappa}{n-1}\,(T_2-T_1)=\frac{n-\kappa}{\kappa-1}\,\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. $\quad(n\neq1)$ |
| **1st LT open system**<br>$q_{12}+w_{t\,12}=h_2-h_1$<br>Rev. pressure change work:<br>$w_{p\,12\,rev}=\int_1^2 v\,\mathrm{d}p$<br>Technical work:<br>$w_{t\,12}=\int_1^2 v\,\mathrm{d}p+j_{12}$ | $q_{12}+w_{t\,12}=h_2-h_1$<br>Isentropic means adiabatic ($q_{12}=0$) and reversible ($j_{12}=0$):<br>$w_{t\,12}=w_{p\,12\,rev}=\kappa\,w_{v\,12\,rev}=h_2-h_1=c_p\,(T_2-T_1)$<br>$w_{p\,12\,rev}=\kappa\,\frac{p_1\,v_1}{\kappa-1}\left[\left(\frac{v_1}{v_2}\right)^{\kappa-1}-1\right]=\kappa\,\frac{R_i\,T_1}{\kappa-1}\left[\left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}}-1\right]=$ etc. | $q_{12}+w_{t\,12}=h_2-h_1=c_p\,(T_2-T_1)$<br>$w_{t\,12}=w_{p\,12}=n\,w_{v\,12}=n\,\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. $\quad(n\neq1)$<br>Rev. SC:<br>$w_{p\,12\,rev}=n\,w_{v\,12\,rev}=\frac{n\,R_i}{n-1}\,(T_2-T_1)=$ etc.<br>$q_{12\,rev}=c_v\,\frac{n-\kappa}{n-1}\,(T_2-T_1)=\frac{n-\kappa}{\kappa-1}\,\frac{p_1\,v_1}{n-1}\left[\left(\frac{v_1}{v_2}\right)^{n-1}-1\right]=$ etc. $\quad(n\neq1)$ |
| **2nd LT / entropy**<br>$\mathrm{d}s=\frac{\mathrm{d}q+\mathrm{d}j}{T}$<br>$\mathrm{d}s=\frac{\mathrm{d}u+p\,\mathrm{d}v}{T}=\frac{\mathrm{d}h-v\,\mathrm{d}p}{T}$ | $\mathrm{d}s=\frac{\mathrm{d}q+\mathrm{d}j}{T}=0\Rightarrow s_2-s_1=0$ | $s_2-s_1=c_v\,\frac{n-\kappa}{n-1}\,\ln\frac{T_2}{T_1}\quad$ (for $n\neq1$)<br>$s_2-s_1=c_v\,\ln\frac{T_2}{T_1}+R_i\,\ln\frac{v_2}{v_1}=c_p\,\ln\frac{T_2}{T_1}-R_i\,\ln\frac{p_2}{p_1}$ |